A nonseparable strong-random extension of the Greguš–Ćirić and Hardy–Rogers fixed-point theorems

Candidate solution packet for arXiv:1706.01634

11 August 2026

Status. Candidate full affirmative answer, likely valid, under the canonical nonseparable formulation using strong (Bochner) random variables and strong random sections. Human verification is requested.

Abstract

Saipara, Kumam, and Cho ask whether two random fixed-point theorems in their paper extend from separable to nonseparable Banach spaces. We give an affirmative extension to every real Banach space after making the standard nonseparable measurability choice: random variables and constant operator sections are strongly measurable. The proof closes a countable family of strong random variables under the operator and rational linear combinations. Fibrewise, its closure is a separable invariant Banach subspace, so the deterministic fixed-point theorem applies. Countably selected approximate residual minimizers converge to the fixed point by explicit residual error bounds, proving its strong measurability. The argument handles both the Greguš–Ćirić condition and the five-term Hardy–Rogers condition.

1 Source question

The source is P. Saipara, P. Kumam, and Y. J. Cho, *Random fixed point theorems for Hardy–Rogers self-random operators with applications to random integral equations*, arXiv:1706.01634v1 (2017), published in *Stochastics* 90 (2018), 297–311 [1]. On printed page 12, after the application theorem, it asks:

> Consequently, $\mathcal{U}(\omega)$ is a random contractive mapping on $Q(\rho)$. Hence, by Theorem 3.2, there exists a random fixed point of $\mathcal{U}(\omega)$, which is the random solution of the equation (4.1). This completes the proof. ∎
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> **Open Problem:** Can Theorems 1.3 and 3.2 be generalized to non-separable Banach spaces?
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> **Acknowledgements**

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Theorem 1.3 is the random Greguš–Ćirić theorem quoted from Saha–Ganguly [2]; Theorem 3.2 is the source paper's random Hardy–Rogers theorem. The source uses separability to identify its several measurability notions. In a nonseparable space those notions separate, so a generalization must say which one is intended.

2 Definitions and the precise extension

Let $(\Omega, \mathcal{F}, \mu)$ be a complete probability space and let X be an arbitrary real Banach space. An X-valued map is *strongly measurable* if it is the almost-sure norm limit of finite-valued simple maps. Write $\mathcal{S}(X)$ for the strong random variables modulo almost-sure equality.

A map $T : \Omega \times X \to X$ is a *strong continuous random operator* if

(S1) $\omega \mapsto T(\omega, x)$ is strongly measurable for every fixed $x \in X$;

(S2) $x \mapsto T(\omega, x)$ is norm-continuous for almost every ω.

This definition agrees with the source's continuous-random-operator convention when X is separable.

We use either of the following almost-sure conditions, imposed for every $\xi, \eta \in \mathcal{S}(X)$.

Greguš–Ćirić condition (GC). There are measurable $a, b, c : \Omega \to [0, \infty)$ with

$$0 < a < 1, \qquad a + b = 1, \qquad c \le \frac{4-a}{8-a} \quad \text{a.s.},$$

such that, abbreviating $T_\omega x = T(\omega, x)$,

$$\begin{aligned} \|T_\omega \xi - T_\omega \eta\| \le a \max\bigl\{ \|\xi - \eta\|, c\bigl(\|\xi - T_\omega \eta\| + \|\eta - T_\omega \xi\|\bigr) \bigr\} \\ + b \max\bigl\{ \|\xi - T_\omega \xi\|, \|\eta - T_\omega \eta\| \bigr\}. \end{aligned} \tag{GC}$$

All values in (GC) are evaluated at the same ω.

Hardy–Rogers condition (HR). There are measurable $\alpha_i : \Omega \to [0, \infty)$ such that $\sum_{i=1}^{5} \alpha_i < 1$ a.s. and

$$\begin{aligned} \|T_\omega \xi - T_\omega \eta\| \le \alpha_1 \|\xi - \eta\| + \alpha_2 \|\xi - T_\omega \xi\| + \alpha_3 \|\eta - T_\omega \eta\| \\ + \alpha_4 \|\xi - T_\omega \eta\| + \alpha_5 \|\eta - T_\omega \xi\|. \end{aligned} \tag{HR}$$

Theorem 1 (Nonseparable strong-random extension)**.** *Let X be any real Banach space and let $T : \Omega \times X \to X$ be a strong continuous random operator. If either* (GC) *or* (HR) *holds for every pair in $\mathcal{S}(X)$, then there is a unique $u \in \mathcal{S}(X)$ such that*

$$T(\omega, u(\omega)) = u(\omega) \qquad \text{for almost every } \omega.$$

Consequently, both theorems named in the source question extend to arbitrary, possibly nonseparable, Banach spaces in the strong-random formulation.

3 Proof intuition

The obstacle is not deterministic fixed-point geometry. It is the loss of a single countable dense subset with which to synchronize null sets and prove measurability. We create the needed countability from the operator itself. Start at zero and repeatedly add all rational linear combinations and all operator images. This produces countably many strong random variables $d_1, d_2, \ldots$. At each outcome, their closed span is separable and invariant under that fibre of T. A countable intersection synchronizes the contractive inequalities on this random subspace, where the deterministic theorem supplies a fixed point.

The fixed point is not selected abstractly. Among the d_k, choose the first one whose residual $\|d_k - Td_k\|$ is below $1/n$. This is a measurable countable choice. Explicit estimates show that small residual forces small distance to the fixed point, so these choices converge strongly to it.

This detour is essential for treating both source theorems uniformly. Raw Picard iteration need not converge in the Greguš–Ćirić case: $T = -I$ satisfies (GC), has the unique fixed point 0, and its nonzero Picard orbits oscillate.

4 Proof

4.1 Strong composition

Lemma 2 (Composition lemma)**.** *If T is a strong continuous random operator and $\xi \in \mathcal{S}(X)$, then*

$$(\mathsf{T}\xi)(\omega) := T(\omega, \xi(\omega))$$

belongs to $\mathcal{S}(X)$.

Proof. Choose finite-valued simple ξ_m with $\xi_m \to \xi$ almost surely in norm. On each member of the finite measurable partition defining ξ_m, $\mathsf{T}\xi_m$ equals a constant section $T(\cdot, x)$; hence $\mathsf{T}\xi_m$ is strongly measurable. Fibre continuity gives $\mathsf{T}\xi_m \to \mathsf{T}\xi$ almost surely. Strong measurability is closed under almost-sure norm limits. □

4.2 A countable random invariant hull

Set $\mathcal{D}_0 = \{0\}$. Recursively let

$$\mathcal{D}_{n+1} = \operatorname{span}_{\mathbb{Q}}\big(\mathcal{D}_n \cup \{\mathsf{T}\xi : \xi \in \mathcal{D}_n\}\big), \qquad \mathcal{D} = \bigcup_{n\geq 0} \mathcal{D}_n. \tag{1}$$

Each $\mathcal{D}_n$ is countable, and Lemma 2 shows that every member is strongly measurable. Enumerate $\mathcal{D}$ as $(d_k)_{k\geq 1}$. For each ω, define

$$Y_\omega = \overline{\{d_k(\omega) : k \geq 1\}}. \tag{2}$$

Because the point set in (2) is rational-linear, Y_ω is a closed real linear subspace of X, hence a separable Banach space. Construction (1) and fibre continuity imply

$$T_\omega(Y_\omega) \subseteq Y_\omega. \tag{3}$$

For every pair (j, k) the assumed random inequality holds almost surely for (d_j, d_k). Intersect these countably many full-measure events with the events on which the coefficient restrictions and fibre continuity hold. On the resulting full-measure set Ω_0, the relevant inequality holds on all pairs $d_j(\omega), d_k(\omega)$ and therefore, by continuity, on all pairs in Y_ω.

If (GC) holds, the deterministic Greguš–Ćirić theorem [3] applied to the closed convex set Y_ω gives a unique fixed point $u(\omega) \in Y_\omega$. If (HR) holds, put $q(\omega) = \sum_i \alpha_i(\omega) < 1$. The right-hand side of (HR) is at most $q(\omega)$ times the maximum of the five usual Čirić distances, so $T_\omega|_{Y_\omega}$ is a quasicontraction. The deterministic quasicontraction theorem [4] again gives a unique fixed point $u(\omega) \in Y_\omega$.

At this stage no measurability of u has been assumed or inferred.

4.3 Residual stability

Lemma 3 (Residual error bounds)**.** *Fix $\omega \in \Omega_0$, suppress it from the notation, and let $u = Tu$ be the fixed point in Y_ω. For $x \in Y_\omega$ set*

$$r = \|x - u\|, \qquad \delta = \|x - Tx\|, \qquad d = \|Tx - u\|.$$

Under (GC),

$$r \leq C_{GC}\delta, \quad C_{GC} := \max\left\{\frac{1+b}{b}, \frac{1-ac+b}{1-2ac}\right\} < \infty. \tag{4}$$

Under (HR),

$$r \leq C_{HR}\delta, \quad C_{HR} := \frac{1-\alpha_5+\alpha_2}{1-\alpha_1-\alpha_4-\alpha_5} < \infty. \tag{5}$$

Proof. In the (GC) case, substituting $y = u$ gives

$$d \le a \max\{r, c(r+d)\} + b\delta, \qquad r \le \delta + d. \tag{6}$$

If the maximum in (6) is r, then $r \le \delta + ar + b\delta$, so $r \le (1+b)\delta/(1-a) = (1+b)\delta/b$. If the maximum is $c(r+d)$, then

$$(1-ac)d \le ac, r + b\delta.$$

Combining this with $r \le \delta + d$ yields $(1-2ac)r \le (1-ac+b)\delta$. The source restriction implies $c \le (4-a)/(8-a) < 1/2$, so both denominators in (4) are positive.

In the (HR) case, substitution of $u = Tu$ gives

$$(1-\alpha_5)d \le (\alpha_1 + \alpha_4)r + \alpha_2\delta.$$

Combining with $r \le \delta + d$ gives

$$(1-\alpha_1-\alpha_4-\alpha_5)r \le (1-\alpha_5+\alpha_2)\delta,$$

which is (5). Its denominator is positive because $\sum_i \alpha_i < 1$. ☐

4.4 Measurable approximation and conclusion

For $k \ge 1$, define the measurable residual

$$\rho_k(\omega) = \|d_k(\omega) - T(\omega, d_k(\omega))\|.$$

It is measurable because both terms are strongly measurable. For each $n \ge 1$, on the event that an index with $\rho_k < 1/n$ exists, let $k_n(\omega)$ be the least such index and set

$$z_n(\omega) = d_{k_n(\omega)}(\omega);$$

set $z_n = 0$ on the complementary event. The least-index events form a countable measurable partition. Consequently z_n is measurable. It is also essentially separably valued: outside one null set, the countable family (d_k) takes values in one separable closed subspace, obtained by closing the span of their countably many essential separable ranges. Thus $z_n \in \mathcal{S}(X)$.

For $\omega \in \Omega_0$, density in (2), continuity of T_ω, and $T_\omega u = u$ imply

$$\inf_k \rho_k(\omega) = 0.$$

Hence $k_n(\omega)$ exists and $\rho_{k_n(\omega)}(\omega) < 1/n$. Lemma 3 gives

$$\|z_n(\omega) - u(\omega)\| \le \frac{C(\omega)}{n} \longrightarrow 0, \tag{7}$$

where $C = C_{GC}$ or C_{HR}. Define $u = 0$ off Ω_0. Equation (7) expresses u as an almost-sure norm limit of strong random variables, so $u \in \mathcal{S}(X)$, and it is a random fixed point.

It remains to prove uniqueness in the random class. Let $u, v \in \mathcal{S}(X)$ be fixed points. Under (GC), the cross term equals $2\|u-v\|$, but $2c < 1$, while both residual terms vanish. Thus

$$\|u - v\| \le a\|u - v\|,$$

so $u = v$ almost surely. Under (HR),

$$\|u - v\| \le (\alpha_1 + \alpha_4 + \alpha_5)\|u - v\|,$$

and the coefficient is below one. This proves Theorem 1. ☐

5 Verification and scope

Checks performed. The proof was audited separately for (i) closure of strong measurability under composition and countable pasting, (ii) synchronization of null sets using only the constructed countable family, (iii) invariance of Y_ω, (iv) positivity of every residual-bound denominator, and (v) uniqueness among strong random fixed points. The included script `code/verify_residual_bounds.py` stress-tested 100,000 admissible scalar configurations for each of (4) and (5); it returned `PASS`. This computation is only a sanity check; Lemma 3 is the proof.

Exact scope. The theorem removes separability of the Banach space completely. It does not assert that the source's bare Borel-section wording is sufficient in every nonseparable codomain. In that setting, Borel measurability is not stable enough for the source's composition steps. Strong measurability is the standard nonseparable replacement and is no extra restriction in the separable setting. The result therefore answers the source's broad generalization question affirmatively while making the indispensable nonseparable measurability convention explicit.

Bounded novelty search. On 11 August 2026 we searched the exact source title and question, its citation neighborhood, and combinations of *Hardy–Rogers*, *nonseparable*, *strongly measurable*, *strong random operator*, *random fixed point*, and *countable invariant subspace*. Guo, Zhang, Wang, and Yuan prove a nonseparable theorem for strong random nonexpansive operators on weakly compact convex sets with normal structure [5]; that result supports the formulation but does not cover the present whole-space Greguš–Ćirić or Hardy–Rogers theorem. No exact answer or the residual-minimizer construction above was located. This is a bounded search, not an exhaustive novelty certification.

Human-review recommendation. Prioritize the countable-invariant-hull step, the use of the deterministic Greguš–Ćirić theorem on Y_ω, and the two residual bounds. Also confirm that the strong-random formulation is accepted as the intended nonseparable reading of the source question.

References

[1] P. Saipara, P. Kumam, and Y. J. Cho, *Random fixed point theorems for Hardy–Rogers self-random operators with applications to random integral equations*, Stochastics 90 (2018), 297–311; arXiv:1706.01634.

[2] M. Saha and A. Ganguly, *Random fixed point theorem on a Čirić-type contractive mapping and its consequence*, Fixed Point Theory Appl. 2012, Article 209 (2012).

[3] L. Ćirić, *On a generalization of a Greguš fixed point theorem*, Czechoslovak Math. J. 50 (2000), 449–458.

[4] L. B. Ćirić, *A generalization of Banach's contraction principle*, Proc. Amer. Math. Soc. 45 (1974), 267–273.

[5] T. Guo, E. Zhang, Y. Wang, and G. Yuan, *L^0-convex compactness and random normal structure in $L^0(\mathcal{F}, B)$*, Acta Math. Sci. Ser. B 40 (2020), 457–469; arXiv:1904.03607.